

Practice 1: Mean-Difference Testing

25117 - Econometrics

Universitat Pompeu Fabra

October 2nd, 2024

Mean Difference Testing

- Mean difference testing is a statistical method used to compare the means of two or more groups or conditions.
- It helps researchers determine whether there are significant differences in the central tendencies of the groups.
- Commonly used in various fields, including social sciences, medicine, and business, to answer research questions.

Question

How can we ensure that groups are comparable?

Hypotheses in Mean Difference Testing

- In mean difference testing, we typically formulate null and alternative hypotheses:
 - Null Hypothesis (H_0): There is no significant difference between the group means.
 - Alternative Hypothesis (H_1 or H_a): There is a significant difference between at least one pair of group means.
- We use sample data to test these hypotheses and make inferences about the population means.
- Several statistical tests are commonly used for mean difference testing:
 - 1 Independent Samples t-test: Compares means of two independent groups.
 - 2 Paired Samples t-test: Compares means of related samples, such as before and after measurements.
 - 3 Analysis of Variance (ANOVA): Compares means of three or more groups.
- The choice of test depends on the study design and research question.

Question

What are the two characteristics required on sample data to draw inference?

Common Mean Difference Tests

1. Independent Samples t-test:

- The independent samples t-test is used when you have two independent groups, and you want to compare the means of a single variable between these groups.
- It assesses whether the difference in means is statistically significant, considering the variability within each group.
- For example, you might use an independent samples t-test to compare the test scores of students who received a specific intervention with a control group of students who did not.
- The test produces a t-statistic and a p-value. A small p-value (< 0.05) suggests a significant difference between group means.

Common Mean Difference Tests

2. Paired Samples t-test:

- The paired samples t-test is used when you have related samples or matched pairs, and you want to compare the means of a variable before and after an intervention or treatment.
- It assesses whether there is a significant change in means within each pair.
- For example, you might use a paired samples t-test to analyze the blood pressure measurements of patients before and after a new medication.
- The test also produces a t-statistic and a p-value. A small p-value indicates a significant mean difference between paired observations.

Common Mean Difference Tests

3. Analysis of Variance (ANOVA):

- ANOVA is used when you have three or more groups, and you want to determine whether there are significant differences in means among these groups.
- It helps identify whether at least one group is different from the others, considering the variability within each group.
- For instance, ANOVA can be applied to compare the performance of students in three different teaching methods (A, B, and C).
- The test provides an F-statistic and a p-value. A low p-value suggests that there is a significant difference in means among the groups.

Steps in Mean Difference Testing

- 1 **Formulate Hypotheses:** Define null and alternative hypotheses based on your research question.
- 2 **Collect Data:** Gather data from your sample groups or conditions.
- 3 **Elaborate an Empirical Strategy:** Select an appropriate mean difference test based on the nature of your data and hypotheses.
- 4 **Conduct the Test:** Perform the statistical test (using software).
- 5 **Analyze Results:** Examine the test statistic, p-value, and effect size to make conclusions.
- 6 **Draw Inferences:** Based on the results, decide whether to accept or reject the null hypothesis and interpret the practical significance.

In Practice (I): COVID-19 and online instruction

- During the COVID-19 pandemic colleges had to adapt rapidly to pandemic-related challenges.
- You are mandated to identify the impact of online-learning on educational outputs.

Questions

- Discuss the pros and cons of online teaching. What would you expect from the theory?
- What would be the gold standard experiment that would allow you to answer this question?
- How would you proceed?
- In practice, what challenges would you face?

In Practice (I): COVID-19 and online instruction

- March 2020 → online instruction until the end of the semester
- Fall 2020 Challenge → should institutions go for online vs. hybrid vs. in-person teaching in the face of uncertainty?
- Institutions and instructors had a lot of discretionary power to decide upon the format

Questions

- There is variation in regarding the instruction format, but why is studying the causal effects of online learning during the pandemic challenging?

West Point RCT: Online vs. In-Person Education

- The West Point RCT aimed to estimate the differences in student outcomes between online and in-person instruction during the COVID-19 pandemic.
- Kofoed et al. (2023) exploited West Point's unique context:
 - West Point students, like their peers at other military academies, have little control over their academic schedules, peer networks, and course assignments.
 - The academy randomly assigns students to academic instructors, class hours, roommates, social networks (called companies), final exam periods, and military mentors.
- Experimental Design:
 - 551 students were randomized across 12 instructors in 38 class sections.
 - The syllabus, graded events, homework assignments, and exams were identical for both online and in-person classes.
 - Both online and in-person class sections were offered for each course hour, and students were randomly assigned within the hour into each modality.

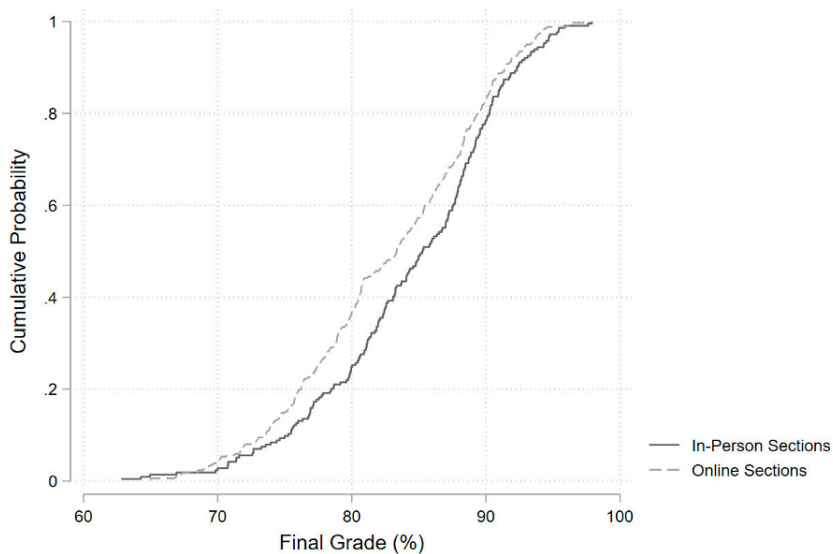
Questions

- With the knowledge you acquired, read, interpret, and discuss the following results.

West Point RCT: Mean-Difference Testing

	Full Sample mean/sd	Online mean/sd	Face to Face mean/sd	Difference b/se
Final Grade (std.)	0.000 (1.000)	-0.092 (1.003)	0.144 (0.980)	0.236*** [0.087]
Final Grade (%)	83.367 (6.998)	82.726 (7.018)	84.376 (6.860)	1.650*** [0.608]
Online	0.612 (0.488)	1.000 (0.000)	0.000 (0.000)	
Female	0.230 (0.422)	0.226 (0.419)	0.238 (0.427)	0.013 [0.037]
Black	0.140 (0.347)	0.139 (0.347)	0.140 (0.348)	0.001 [0.030]
Hispanic	0.033 (0.178)	0.027 (0.161)	0.042 (0.201)	0.015 [0.016]
Asian	0.056 (0.231)	0.050 (0.219)	0.065 (0.248)	0.015 [0.020]
NCAA Athlete	0.290 (0.454)	0.267 (0.443)	0.327 (0.470)	0.060 [0.040]
Prep School	0.142 (0.349)	0.142 (0.350)	0.140 (0.348)	-0.002 [0.031]
Prior Enlisted	0.169 (0.375)	0.160 (0.367)	0.182 (0.387)	0.022 [0.033]
CEER Score	623.806 (73.377)	623.083 (70.981)	624.944 (77.155)	1.861 [6.419]
Observations	551	337	214	551

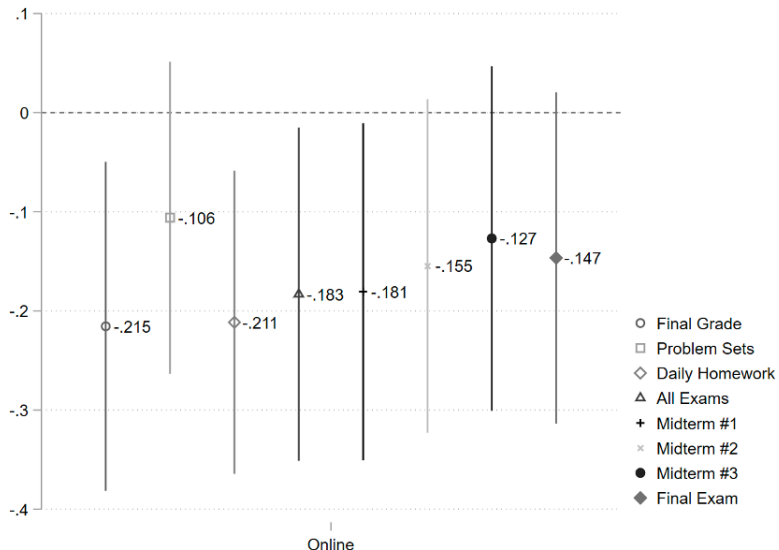
West Point RCT: Mean-Difference Testing



West Point RCT: Mean-Difference Testing

	(1)	(2)	(3)	(4)	(5)	(6)
		Final Grade				
	Overall	Q.10	Q.25	Q.50	Q.75	Q.90
Online	-0.215** (0.084)	-0.090 (0.173)	-0.226** (0.111)	-0.351*** (0.119)	-0.115 (0.091)	-0.194* (0.112)
Instructor FE	Yes	Yes	Yes	Yes	Yes	Yes
Class Day FEs	Yes	Yes	Yes	Yes	Yes	Yes
Time of Day FEs	Yes	Yes	Yes	Yes	Yes	Yes
Exog. Controls	Yes	Yes	Yes	Yes	Yes	Yes
Observations	551	551	551	551	551	551
R^2	0.173	0.126	0.127	0.130	0.103	0.098
Robust SEs p-values	0.011	0.603	0.042	0.003	0.209	0.084
Wild Bootstrapped SEs p-values	0.013	0.009	0.016	0.018	0.014	0.009

West Point RCT: Mean-Difference Testing



In Practice (II): UI benefits vs Unemployment

- Are Unemployment Insurance benefits distortionary (i.e., generating behavioral responses)?
- You are mandated to identify the impacts of cash bonuses on the duration of insured unemployment

Questions

- Discuss how unemployment benefits might or might not yield longer unemployment duration
- What would be the gold standard experiment that would allow you to answer this question?
- How would you proceed?
- In practice, what challenges would you face?

In Practice (II): UI benefits vs Unemployment

- **Woodbury and Spiegelman (1987):**
 - Between mid-1984 and mid-1985, the Illinois Department of Employment Security conducted two RCTs.
 - Objective: Test the effectiveness of cash bonuses in reducing the duration of insured unemployment.
- **Background:**
 - Lack of clear understanding of how UI affects recipient behavior.
 - UI benefits may act as a subsidy to additional job search.
 - UI benefits may subsidize leisure rather than job search.
- **Key Question:** Do monetary incentives influence job search behavior (i.e., are UI transfers distortionary)?
- **Treatment Description:**
 - Two treatments with a \$500 bonus payment (i.e., approx. 5% of annual wage and salary payments!) - one to claimants and one to employers.

UI benefits vs Unemployment

TABLE 2—CHARACTERISTICS OF CLAIMANTS ASSIGNED TO EXPERIMENTAL GROUPS

	Control		Claimant Experiment		Employer Experiment	
	<i>N</i>	Proportion	<i>N</i>	Proportion	<i>N</i>	Proportion
Total	3,952	1.000	4,186	1.000	3,963	1.000
Male	2,162	0.547	2,357	0.563	2,131	0.538
White	2,497	0.632	2,723	0.651	2,565	0.647
Black	1,072	0.271	1,050	0.251	1,014	0.256
Hispanic, Native American, Other	383	0.097	413	0.099	384	0.097
Age 20–29	1,680	0.425	1,827	0.436	1,679	0.424
Age 30–39	1,315	0.333	1,357	0.324	1,292	0.326
Age 40–49	708	0.179	776	0.185	740	0.187
Age 50–54	248	0.063	226	0.054	252	0.064
Weekly Benefit Amount:						
\$51	347	0.088	355	0.085	333	0.084
\$52–\$90	794	0.201	887	0.212	861	0.217
\$91–\$120	666	0.169	738	0.176	711	0.179
\$121–\$160	749	0.190	822	0.196	716	0.181
\$161	1,396	0.353	1,384	0.331	1,342	0.339
Dependents' Allowance	1,834	0.323	1,955	0.345	1,883	0.332

UI benefits vs Unemployment

TABLE 3—MEANS OF PROGRAM VARIABLES BY EXPERIMENTAL GROUP^a

	Control		Claimant Experiment		Employer Experiment	
	Mean	SE of Mean	Mean	SE of Mean	Mean	SE of Mean
Benefits Paid (\$):						
1) State Regular, First Spell	2,267	27.5	2,074	26.7	2,159	27.4
2) Total, First Spell	2,558	33.8	2,329	32.9	2,446	33.8
3) State Regular, Benefit Year	2,487	27.0	2,328	26.3	2,426	27.0
4) Total, Benefit Year	2,786	33.1	2,592	32.2	2,725	33.8
Weeks of Insured Unemployment:						
1) First Spell	18.3	0.205	17.0	0.199	17.7	0.205
2) Benefit Year	20.1	0.194	18.9	0.188	19.7	0.194
	Propor- tion	SE of Pro- portion	Propor- tion	SE of Pro- portion	Propor- tion	SE of Pro- portion
Proportion of Claimants Who:						
1) Exhausted Benefits	0.478	0.008	0.446	0.008	0.464	0.008
2) Ended Benefits within 11 weeks	0.353	0.008	0.408	0.008	0.384	0.008
N	3,952		4,186		3,963	

UI benefits vs Unemployment

TABLE 4—DIFFERENCES BETWEEN CONTROL GROUP AND EXPERIMENTAL GROUP MEANS

	Claimant Experiment minus Control		Employer Experiment minus Control	
	Difference of Means	SE	Difference of Means	SE
Benefits Paid (\$):				
1) State Regular, First Spell	-193 ^a	38.30	-108 ^a	38.82
2) Total, First Spell	-229 ^a	47.15	-112 ^b	47.79
3) State Regular, Benefit Year	-158 ^a	37.70	-61	38.21
4) Total, Benefit Year	-194 ^a	47.17	-61	46.80
Weeks of Insured Unemployment:				
1) First Spell	-1.37 ^a	0.29	-0.67 ^b	0.29
2) Benefit Year	-1.15 ^a	0.29	-0.36	0.27
	Difference of Proportions		Difference of Proportions	
	Difference of Proportions	SE	Difference of Proportions	SE
Proportion of Claimants Who:				
1) Exhausted Benefits	-0.032 ^a	0.011	-0.014	0.011
2) Ended Benefits within 11 Weeks	+0.055 ^a	0.011	+0.031 ^a	0.011